

EX 6

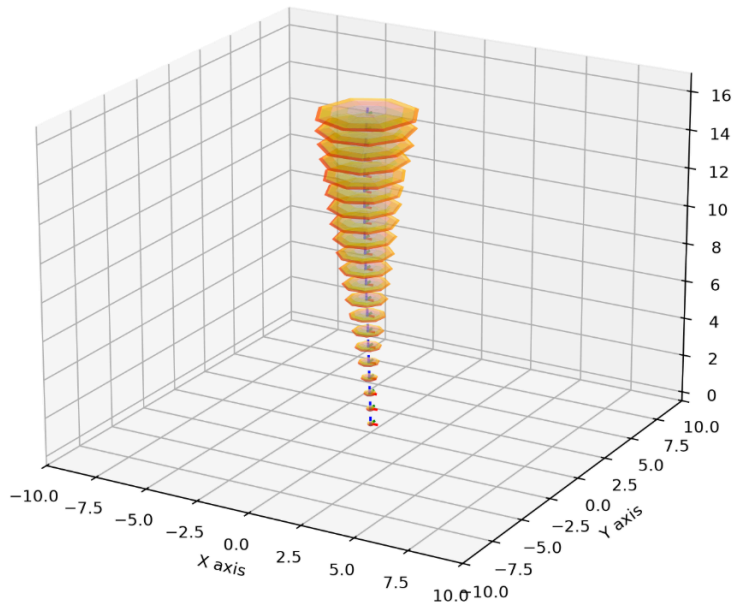
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[Github](#)

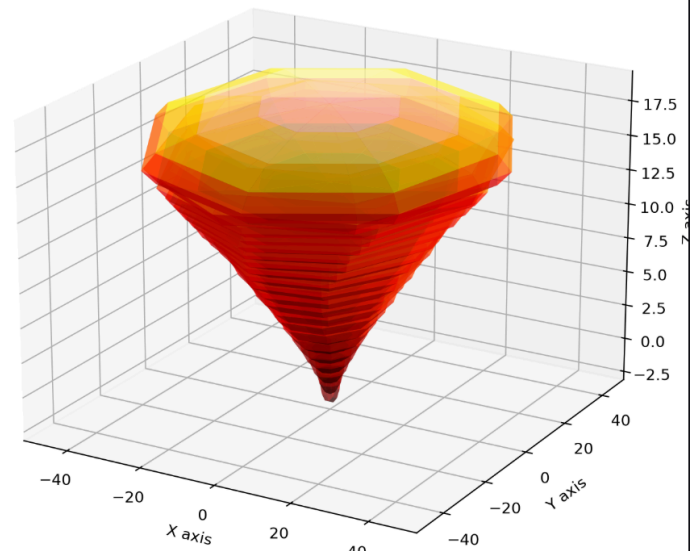
6.1:

As we can see, covariance of the bundle are very sensitive to the prior noise:

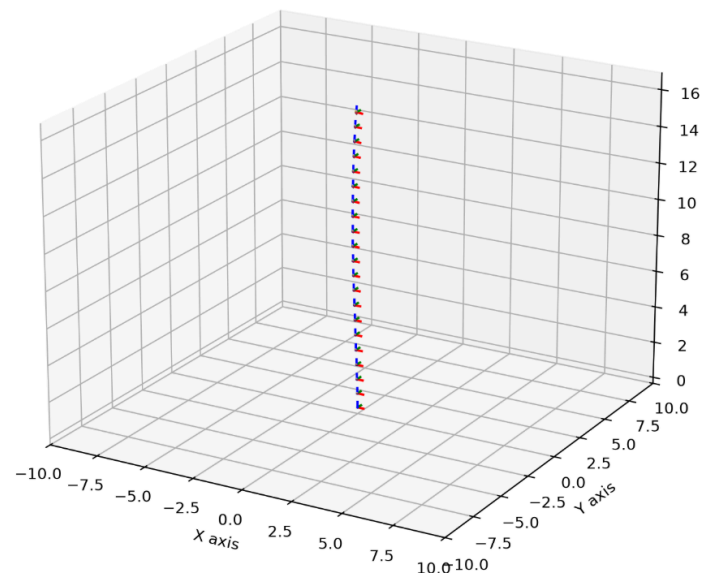
6.1 - Bundle 0→20 | prior noise: $I \times 0.05$ ($\sigma=0.05$)
Frame locations with marginal covariances



6.1 - Bundle 0→20 | prior noise: Unit matrix ($\sigma=1.0$)
Frame locations with marginal covariances



6.1 - Bundle 0→20 | prior noise: $I \times 1e-6$ ($\sigma=1e-6$)
Frame locations with marginal covariances



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--- Relative pose & covariance for c0=0 → c_k=20 ---
Graph size before optimization: 997
Initial estimate size: 299
Window 0→20: factors=997, error before=1476.51, after=221.35

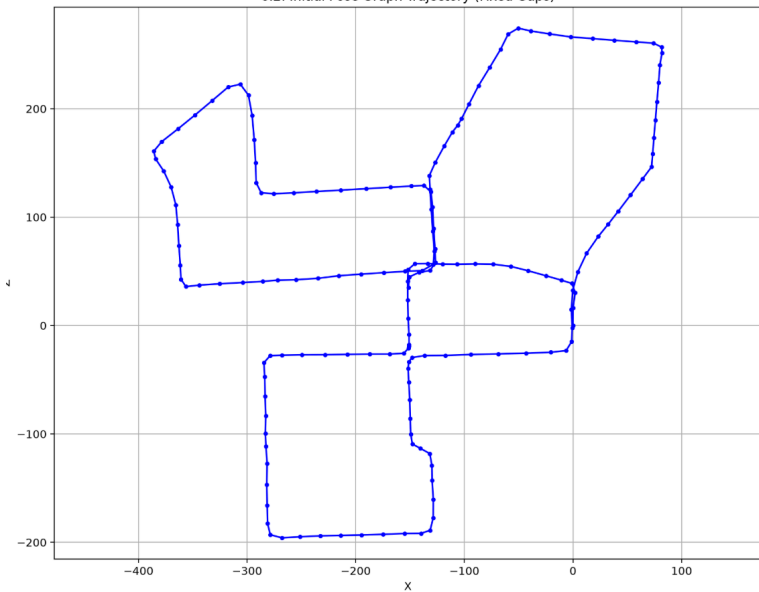
Relative Pose between keyframes c0 and c20:
R: [
    0.996866, 0.0363718, 0.0702476;
    -0.0362185, 0.999338, -0.0034542;
    -0.0703267, 0.000899111, 0.997524
]
t: 0.160889 -0.342595 15.9539

Conditional covariance P(c20 | c0):
[[ 0.00000486 -0.00000108 0.00000964 0.0000206 0.00021087 -0.00000526]
 [-0.00000108 0.00000391 -0.00000534 -0.00017572 -0.00005956 0.00018967]
 [ 0.00000964 -0.00000534 0.00003502 0.00014524 0.00042007 -0.00007597]
 [ 0.0000206 -0.00017572 0.00014524 0.00873532 0.00149945 -0.00949423]
 [ 0.00021087 -0.00005956 0.00042007 0.00149945 0.00995507 -0.00204645]
 [-0.00000526 0.00018967 -0.00007597 -0.00949423 -0.00204645 0.02759951]]

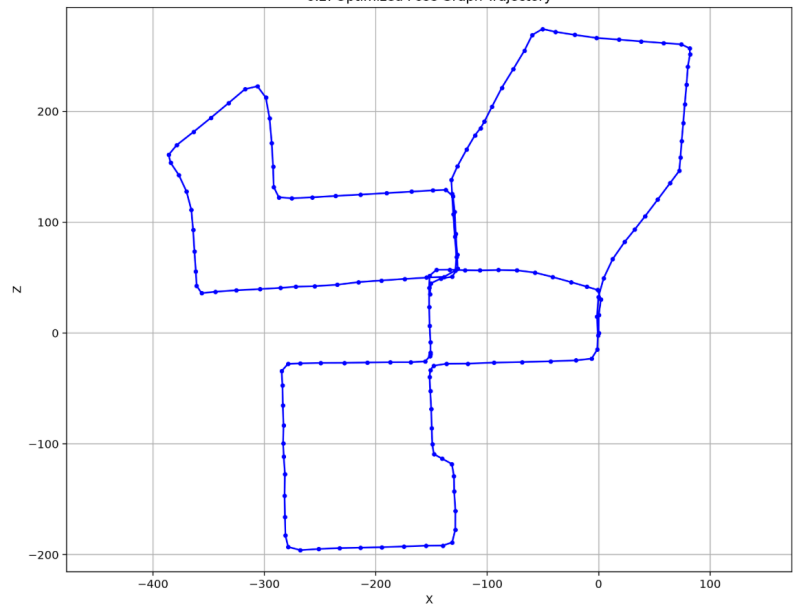
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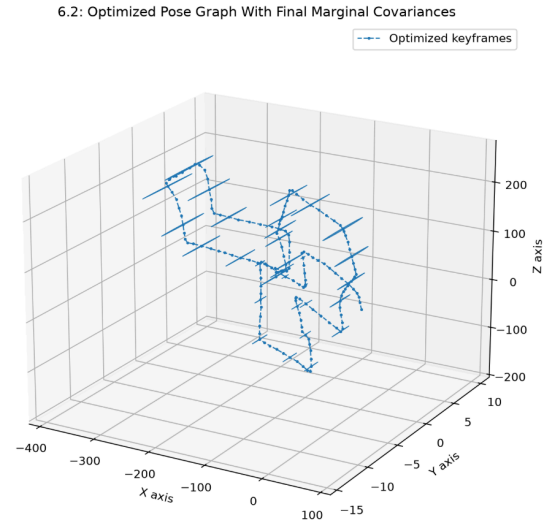
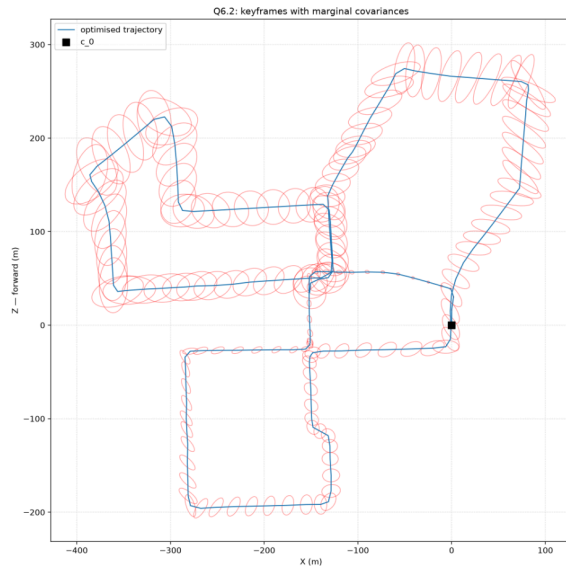
6.2:

6.2: Initial Pose Graph Trajectory (Fixed Gaps)



6.2: Optimized Pose Graph Trajectory





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Pose graph error BEFORE optimization: 1.15400778e-16
Pose graph error AFTER optimization: 2.79138345e-21
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A reasonable initialization for the pose graph is obtained by chaining the relative pose constraints estimated from the local bundle-adjustment windows. We set the first keyframe pose to the identity, and then compose each consecutive relative motion with the previous pose. This initialization is natural because the pose graph is built directly from these relative measurements, so the chained trajectory is already consistent with the available odometry constraints.

The initial pose graph trajectory is shown in the first plot. After optimizing the pose graph, the resulting trajectory is visually almost identical to the initial trajectory. This is expected, because the graph contains mainly consecutive relative-pose factors and a prior on the first keyframe, without loop-closure constraints or additional global measurements. Therefore, the initial estimate already satisfies the between-factors very well.

The factor graph error before optimization was approximately zero, and the error after optimization was also approximately zero. These values are not surprising: since the initialization was constructed by composing the same relative motions used as pose-graph constraints, the residuals of the between-factors are already nearly

zero before optimization. As a result, the optimizer has almost no correction to apply to the keyframe locations.

The final marginal covariances are shown in the third plot. The covariance ellipsoids represent the positional uncertainty of the optimized keyframes. Since the pose graph is essentially a chain of relative constraints, uncertainty accumulates along the trajectory as we move farther from the anchored first pose. The elongated ellipsoids indicate that the uncertainty is anisotropic, with larger uncertainty along certain motion directions. This behavior is reasonable for a visual-odometry-based pose graph without loop closures, where drift is not globally corrected.

Note:

We transitioned from ORB to AKAZE because the AKAZE descriptors yielded superior tracking accuracy.